



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 01

Experiment Title: Chip-Design (Full Adder)

Submitted by:

Name: Md. Akif Hossain

Student ID: 22201029

Section: C1

Submitted to:

Md. Rasheduzzaman

Lecturer

Department of Computer Science and Engineering

Date of Submission: 27/9/26

1. Title of the Experiment

Design and Simulation of a 1-Bit Full Adder

2. Objectives

This experiment's primary goals are:

- to comprehend the fundamental operation of a 1-bit Full Adder.
- to use Verilog HDL to create a circuit for a full adder.
- to comprehend how Sum and Carry outputs work.
- to use Xilinx ISE to implement the design.
- to use ISim to simulate the circuit.
- to confirm the result for every conceivable combination of inputs.

A, B, and Cin are the three inputs of a full adder. Sum and Cout are the two outputs it generates.

3. Software and Tools Used

Hardware Needs

- Computer/Laptop
- At least 4 GB of RAM
- Mouse and Keyboard

Software Requirements

- Verilog HDL Simulator Xilinx ISE 14.7
- VirtualBox (if a virtual computer is used to run Xilinx ISE)
- Verilog HDL was used to create the Full Adder, and Xilinx ISE's ISim Simulator was used to simulate it.

4. Design Methodology

A 1-bit Full Adder has three inputs:

- **A** – First input
- **B** – Second input
- **Cin** – Carry input

It has two outputs:

- **Sum**
- **Cout** – Carry output

The logic equations of the Full Adder are:

$$\text{Sum} = A \oplus B \oplus \text{Cin}$$

$$\text{Cout} = AB + \text{BCin} + \text{ACin}$$

The RTL schematic generated by Xilinx ISE represents the internal logic of the Full Adder.

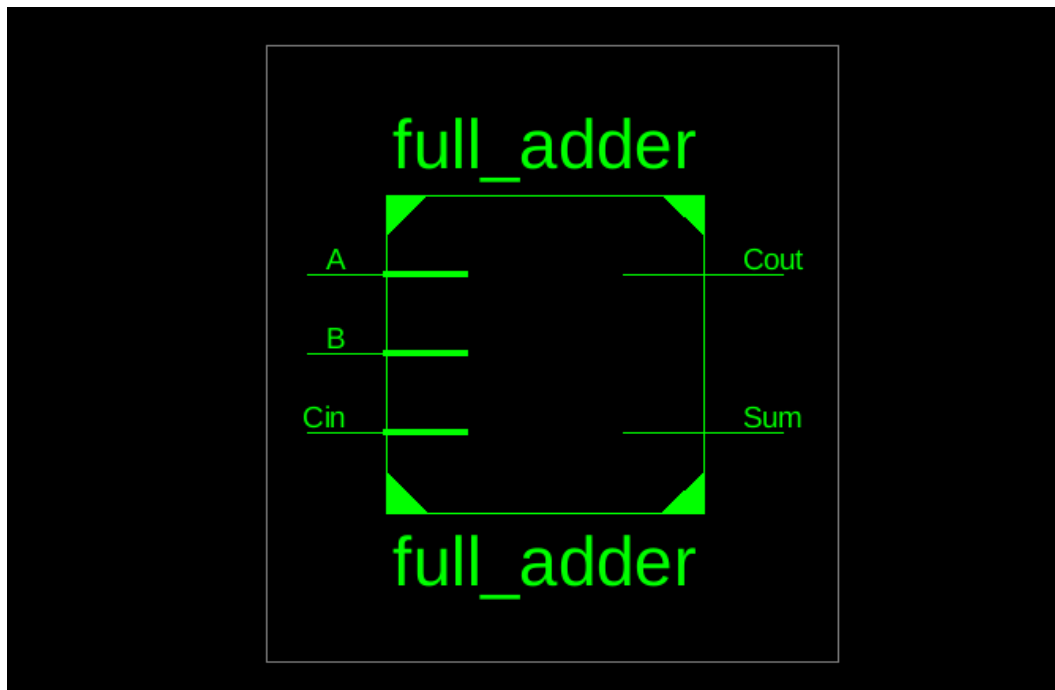


Fig: RTL schematic diagram

6. Output Analysis / Observations

The 1-bit Full Adder was simulated using ISim. All possible combinations of the three inputs A, B, and Cin were applied during the simulation.

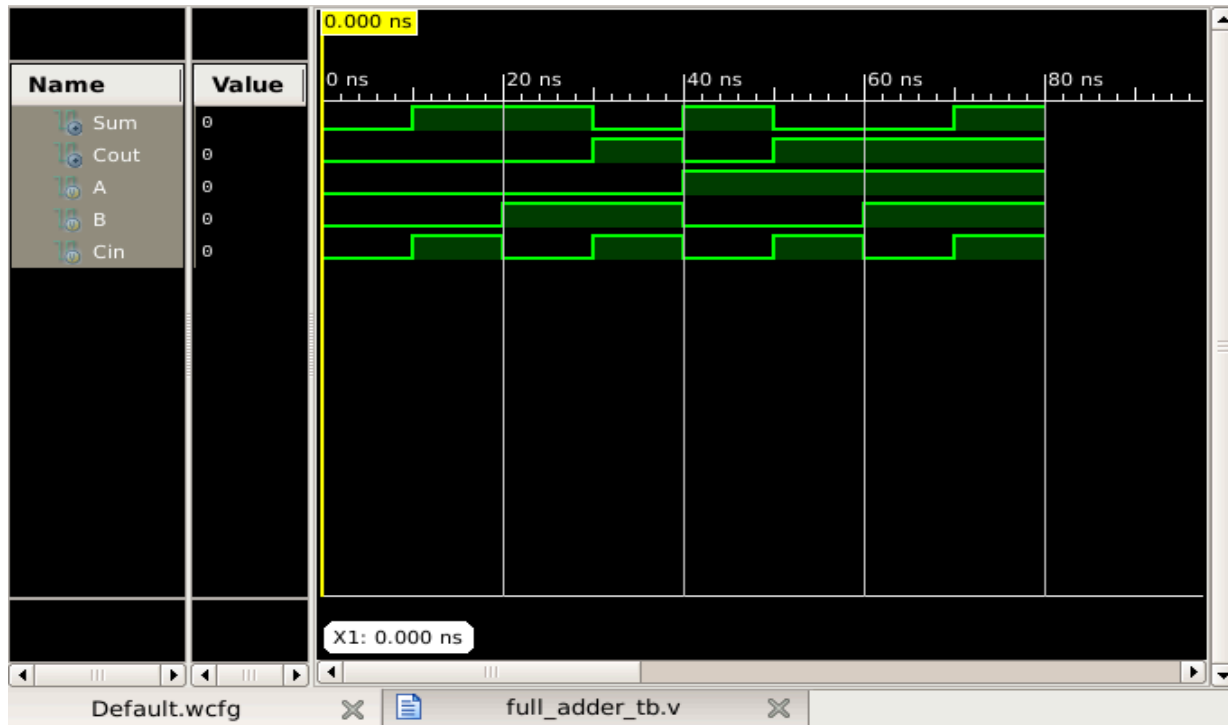
Since there are three binary inputs, there are 8 possible input combinations.

The expected outputs are:

A	B	Cin	Sum	Cout
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0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The ISim waveform showed that the Sum and Cout outputs changed correctly according to the input values.



The observed simulation output matched the expected truth table.

7. Result

Verilog HDL was used to successfully design the 1-bit Full Adder, and Xilinx ISE and ISim were used to model it. For every feasible combination of inputs, the simulation generated the right Sum and Carry-out.

As a result, the experiment's goal was effectively accomplished.

8. Github Repository Commits

GitHub Repository:

<https://github.com/Akifhossain29/ISE-14.7-VLSI>

GitHub Commit Link:

<https://github.com/Akifhossain29/ISE-14.7-VLSI/commits/main/>

The repository contains the Verilog source code and related files used for this experiment.

9. Conclusion

This experiment taught me how to use Verilog HDL to create a 1-bit Full Adder. Additionally, I learnt how to use ISim simulation to confirm the design's functionality and implement it in Xilinx ISE.

The simulation made it easier to comprehend how three binary inputs are used to generate Sum and Carry-out. The experiment was effectively finished, and the anticipated outcomes were attained.